NIAGARA 4 · API COMPARISON

glineRestApi v2
How it compares

Built-in XML API · **glineRestApi v2** · Third-party connector · Built-in REST · Analytics

Side-by-side comparison of approaches to interact with Niagara N4 stations — from the perspective of the **v2** edition.

All v2 figures verified live on a Niagara 4.14 station.

Documented version **v2.4.0** (module `glineRestApi-rt`) · SHA-256 `efafe1e6…f3744f833`

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FEATURE MATRIX

Feature comparison by channel

Feature	glineRestApi v2	Built-in XML	Third-party conn.	Built-in REST	Analytics
Read values	Yes	Yes	Yes	Yes	Read-only
Write values	Yes	Yes	Yes	Yes	No
Create / delete components	Yes	No	Yes	Limited	No
Deep copy (link rebuild)	Yes	No	No	No	No
Rename (link-safe)	Yes	No	No	No	No
Link / unlink	Yes	Partial	Yes	No	No
Invoke actions	Yes	Partial	Yes	No	No
Atomic batch (single POST)	11 ops + BQL	No	Limited	No	No
Facet editing (units, enum ranges)	Yes	No	No	No	No
BQL query → JSON	Yes	No	No	No	Yes
Program management	Yes	No	No	No	No
Alarm/history extension setup	Yes	No	No	No	No
Built-in auth (Basic)	Yes	Yes	Yes	Yes	Yes
CIDR allow-list	Yes	No	Partial	No	No
Alarm / history query	Via Web Box	Yes (best)	Partial	Partial	Yes
License cost	On request	Free	Per station	Free	Subscription
SMA required	No	No	No	Yes	No
Port	8082	—	—	—	—

v2 is the authenticated edition, distributed on request. It can run side by side with v1.

WHAT V2 ADDS

v2 = v1 + security + query + editing

BQL query → JSON

- Run BQL over HTTP, get a structured table
- `{"columns":[…],"rows":[…],"count":N}`
- Ready for reports, dashboards, AI pipelines

Program management

- `POST /api/program/write` — write source
- `POST /api/program/addslot` — add slot
- `GET /api/program/status` — read status

Facet editing (v2.4)

- Numeric: `min` / `max` / `precision`
- `units`, enum `range=Low,Mid,High`
- Read a specific slot's facets back

Built-in authentication

- HTTP Basic Auth, configurable at runtime
- Missing or wrong credentials → **401**
- CIDR allow-list → **403** outside the list

NEW · v2

Alarm & history extension setup over REST

Extensions are ordinary components — v2 can attach, configure, and **activate** them over HTTP. No Workbench required.

```
curl -u "<USER>:<PASS>" -X POST http://<HOST>:8082/api/batch \
  -H "Content-Type: application/json" -d '[
{"OP":"add","PATH":"/myFolder/pt","T":"history:NumericCovHistoryExt","N":"he"},
{"OP":"mod","PATH":"/myFolder/pt/he","N":"enabled","V":true},
{"OP":"add","PATH":"/myFolder/pt","T":"alarm:AlarmSourceExt","N":"ae"},
{"OP":"mod","PATH":"/myFolder/pt/ae","N":"alarmClass","V":"/AlarmClass"}
]'
# → enabled=true, active=true — history collection is live
```

■ **Reading the data:** the REST API has no alarm/history *query* endpoint by design — read it through the station's **Web Box** feature (`/wsbox/query/data/<ord>`), also automatable.

ENDPOINTS

v2 HTTP endpoints

Endpoint	Method	Edition	Purpose
`/api/status`	GET	**v2**	Service status (auth-exempt liveness probe)
`/api/config`	GET	**v2**	Browse the station tree
`/api/batch`	POST	**v2**	11 batch ops + BQL + facet
`/api/program/write`	POST	**v2**	Write program source
`/api/program/addslot`	POST	**v2**	Add a program slot
`/api/program/status`	GET	**v2**	Read program status
`/api/program/compile`	POST	**v2**	Compile the program

`/api/modules` and `/api/types` exist in v2 but are informational stubs (fixed hints) — use `/api/config` or BQL to enumerate.

Example — BQL to JSON

```
curl -u "<USER>:<PASS>" -X POST http://<HOST>:8082/api/batch \
  -H "Content-Type: application/json" \
  -d '[{"OP":"bql","Q":"select name from control:NumericWritable"}]'
```

DECISION GUIDE

When to use v2

Choose v2 when…

- You need authentication and access control
- You need BQL results as JSON tables
- You need program management over HTTP
- You need facet editing (units, enum ranges)
- You need alarm/history extension setup over REST

Stay on v1 when…

- You are on a trusted, internal network
- You want a free, single-.jar drop-in
- You only need batch component operations
- You need real `/api/modules` and `/api/types` listings

v2 = v1 + authentication + access control + BQL + program management + facet editing + alarm/history extension automation.

CONTACT

Let's build your automation

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All glineRestApi v2 figures verified live on a Niagara 4.14 station. Features, availability and pricing of third-party products may vary — verify directly with each vendor.

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